title: "161.304 Project - Risk factors associated to Bovine Fasciolosis in Brazil script"
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Introduction
Bovine fasciolosis is a parasitic disease caused by Fasciola hepatica, the common river fluke, causes economic damages by creating unsalable beef. Thus, understanding and predicting the disease has value. In 2015 Silva et al created a multivariate linear model to predict the disease positivity index (PI) in the state of Rio Grande do Sul in Brazil and the municipality level by year. Their model, excluded data from 2008 and 2009 as the data was more extreme due to La Nina. This paper creates two causal models to predict PI using the same data for municipalities year on year, but including La Nina data to handle a complex model. Comparing models using quadratic approximation which are efficient on large data sets, and Markov chain Monte Carlo algorithms which are robust for complex models such as the fasciolosis model.

The 2015 model, shown below, used the variables log of PI from the prior year, prior year winter maximum temperature and altitude. The two models in this paper use prior year winter maximum temperature and altitude, and the second adds summer rain, percentage of water and La Nina years for each municipality.

2015 paper model.
$$
Pi_general = e^{2.736 + 0.401 * log(PI.yearBefore) - 0.001*Ave.Alt - 0.444
T_{max}WinterYearBefore} -1
$$

Their model was chosen by selecting the model with the highest R^2 and least number of predictors and had an accuracy between 48% and 66% depending on the year. Their approach was to choose a model from the data, rather than understanding the natural phenomenon generating the data. Resulting in a poor fit.

the phenomenon of fasciolosis.
Fasciola hepatica's eggs develop on a moist substrate. Hatched larvae must find a snail host within eight hours. Wherein they progress to metacercariae, are passed and attach to vegetation until ingested by a mammal. From inside the mammal they produce eggs which are passed.

The cycle takes roughly three to four months at the quickest (NCEZIDI, 2019). With most infection occurring in the wet season Kipyegen et al (2022). December to May in Brazil. In order to thrive fasciola require humidity of around 60% and temperature between 15^oC, and 25 ^oC. Below 15 the parasite dies. Above 25 their development slows.

Methods and Analysis
Analysis of variables.
According to theory, where infection spikes following the wet season, a warm winter and adequate summer rain are necessary. These variables (TempMaxWinYearBefore and PrecipSumYear) were sufficient climatic predictors after testing the others for interaction (appendix 1). water on ground or from rain, vegetation and humidity are required. Humidity is affected by altitude. During the test the interactions between variables were explored. The below plots show these with the chosen predictors used in the DAG.

As altitude increases, PI decreases. Dryness, fewer farms and less water cause this. Water runs to a local minima, higher altitudes tend to have less water. Altitude implies rain patterns; rain shadows and coastal rain currents. Low lands are usually coastal and coasts get more rain.

Most observations occur where water is 15% to 20%. The Lymnaeidae snail host favours ponds and reservoirs. Above 20% water PI decreases. Again, more water cover, less farms.

Minimum temperature is not a concern as if it gets too cold (<15^oC), the worm dies (steward, 1999). Above 20^oC development slows, less infection, less PI.
Vegetation becomes scare the higher one climbs as environmental stress reduces the types and quantity of plants. Mammalian hosts eat roughage to get infected. Too much cover (trees) means less pasture and less PI.
La Nina is a dry period leading to drought in do Sul. Rain and water cover is reduced as it evaporates. Despite this there is no clear difference between La Nina years and max winter temperature or vegetation; only reduced summer rain. Which affects PI due to most municipalities having 60 in rain (drier). In regular years when the rain is heavy, the fluke can't successfully infect many hosts. They drown or are washed away. La Nina years typically have PI less than 0.25, while regular years cover the whole range from 0 to 100%.

DAG
The following DAG was identified with theory and testing for causal paths (appendix 2 & 3).

Accordingly, to predict PI, the causal model needs to:
- Condition on temperature in order to close the back doors from PI to rain via altitude and water.
- Include Altitude to close forks into water, rain and PI.
- Not include vegetation as it is a collider and is closed by default.
- Not include water if the total effect of altitude is of interest.
- Not include rain if the total effect of altitude is of interest.

To predict PI with La Nina, the model needs to:
- include La Nina and rain.
- Include altitude to close back doors and forks into rain. Making the pipe open from rain through water to PI
- include temperature in order to close the back door into rain.

Models
The following causal predictive model without La Nina was created, including the total effect of altitude on PI and closing the back door through temperature by conditioning on it. It is similar tp the 2015 model, omitting only PI from the previous year. .
$$
PI \sim \mathcal{Normal}(\mu_i, \space \sigma) \\
\mu_i = \alpha + \beta_A Altitude_i + \beta_T Temperature_i \\
\alpha \sim \mathcal{Normal}(0, \space 0.5) \space\ \\
\beta_A \sim \mathcal{Normal}(0, \space 0.25) \\
\beta_T \sim \mathcal{Normal}(0, \space 0.25) \\
\sigma \sim \mathcal{Exponential}(1) \\
$$

Predicting PI with La Nina requires an interaction between La Nina, PI and rain. In this mode La Nina is coded as a factor (regular = 1, La Nina year = 2). It requires including rain, water and temperature in order close forks, but does not require vegetation as it is a collider.

$$
PI \sim \mathcal{Normal}(\mu_i, \space \sigma) \\
\mu_i = \alpha_{La Nina[k]} + \beta_A Altitude_i + \beta_W Water_i + \beta_{R[k]}

Rain_{i_{LaNina[k]}} + \beta_T Temperature_i \\
\alpha_{La Nina[k]} \sim \mathcal{Normal}(0, \space 0.2) \space\ for \space k= 1..2\\
\beta_A \sim \mathcal{Normal}(0, \space 0.2) \\
\beta_W \sim \mathcal{Normal}(0, \space 0.2) \\
\beta_{R[k]} \sim \mathcal{Normal}(0, \space 0.2) \space\ for \space k= 1..2\\
\beta_T \sim \mathcal{Normal}(0, \space 0.2) \\
\sigma \sim \mathcal{Exponential}(1) \\
$$

To represent the paper's predictive model, this one is used:
$$
PI \sim \mathcal{Normal}(\mu_i, \space \sigma) \\
\mu_i = \alpha + \beta_A Altitude_i + \beta_T Temperature_i + \beta_P PIYearBefore_i\\
\alpha \sim \mathcal{Normal}(0, \space 0.2) \space\ \\
\beta_A \sim \mathcal{Normal}(0, \space 0.2) \\
\beta_T \sim \mathcal{Normal}(0, \space 0.2) \\
\beta_P \sim \mathcal{Normal}(0, \space 0.1) \\
\sigma \sim \mathcal{Exponential}(1) \\
$$
The exact paper model has a very bad fit, adjusted $R^2 = 0.31 (appendix 4). To compare the models on like terms, all models are fit quadratic approximation and MCMC.

For all model analyses, variables are z-score standardised. Null values and PI = 0 values dropped to reduce skew and enable fitting.

Models are fitted twice in order to compare the quadratic approximation (QA) and MCMC algorithms. Models are compared for out of sample fit using WAIC and PSIS scores. Quadratic approximation is computationally efficient which has benefits on the large dataset being used. It uses local estimation and can get 'stuck' if priors are too flat because it relies on prior curvature. While more computationally intensive, MCMC is more robust and can deal with complex models well, like the La Nina model. MCMC's global estimation can also get stuck on flat priors. For each model informative priors were selected and checked alongside the posteriors (appendix 5).

Results
PI_yearBefore widens the CI on the intercept, anticipating more variance. Aside from this, the paper model and the second, causal model are identical.

Altitude in the La Nina model has a lessened impact on PI, because the model also includes water and rain, which are forks with Altitude include some of the causal signal of Altitude. The total effect of altitude is reduced when they are present.
It is also evident that in a La Nina years the variance on rain is greatly increased (bR[2]), from 0 to about -0.3. The intercept also shifts to 0.5 in La Nina years with the CI crossing zero on the left. Consistent with the expectation that La Nina creates more extreme data points. Seen below in the plot and summary tables.

Comparing model prediction
The paper and causal models look to have similar actual vs precipitated PI values. Both predict data points frequently below the minimum value for real PI. Seldom as high as the actual data. Most of the real and predicted data is in the models' 89% CI. This being because most PI values in the real data are at -1 SD from the mean, and the models' CI encompass this range. Even with informative priors, the fit is generating more extreme values left of the mean (low PI) than the real data while

not being able to predict extreme value to the right. The predicted values appear normally distributed about the mean, which is due to the normally distributed priors.

The La Nina model has a much more variable credible interval, but is similar to the other model for the same reasons.

This normal distribution of predicted values can be seen in the following plot. The reason the QA models perform this way is due to their local estimation algorithms being fed normal priors. The models expect normality.

Comparatively, the QA models all have similar means of roughly 0, as expected with the priors. Overall a poor fit even though the real data is mostly within the credible interval. The models are under fitting and not learning enough from the data.

Comparing monte carlo models

After fitting, MCMC models were analysed and all converged (appendix 5).The paper and causal models now have quite different mean lines. The paper model varies much more, perhaps due to the PI_yearBefore variable. Neither is a good fit, still reliably predicting normally distributed PI values. The La Nina model is very accurate. Predicting similar patterns to the data within a wider range; between -3 SD and +6.5 SD where the real data will range from -1SD to +6 SD.

Unlike the QA models, the predicted densities are varied. The paper model has become more normal, causal has not changed and the La Nina model is very skewed, matching more closely the actual data.

Observing the precis outputs for each model, the estimates are the same as the QA models, at least to two decimal places. Again, we see all models converged with an Rhat of 1. The mean of the models are still zero.

Comparing all model predictions:

The best models for prediction for out of model predictions is the La Nina model, according to WAIC the MCMC model is better, and for PSIS the QA model. Under PSIS the La Nina models perform better because we have a known process and a complex model with many influential outliers. PSIS is a better criterion in this case. With a dSE less than 1, often less than 0.5 (random variation in the code sometimes changes this). WAIC tends to favour the QA La Nina model despite penalising more complex models. The paper models are both worse at predicting out of sample, scoring the lowest in both criterion. Due to the presence of the PI_yearBefore variable. See the table and graphs below.

discussion and conclusion

Silva et al found that PI-YearBefore was the most important variable in their model. This paper finds it detrimental. They reasoned that PI in one year led to PI in the next. This isn't true, because PI is not a measure of the infected population, but a feature of having infected and uninfected individuals cohabiting and some number of them being tested. In the raw data a PI of 50% is achieved where 2 cows were slaughtered and a PI of 100% where only one cow was slaughtered. In the raw data, what tends to happen is PI is lower, where more cows are slaughtered. While observed infection increases proportional to the number of cows slaughtered up until high numbers are slaughtered where the relationship begins to vary much more significantly. PI is therefore a misleading measure as it makes infection seem high in areas were little testing was done.

If the conditions for infection persist year on year, then infection can happen yearly. Even as the PI score fluctuates. PI doesn't tell anyone how many infected there are, only what portion of those

tested are infected.

Perhaps the most important outcome of this paper is that a causal model including the La Nina interaction, something not attempted by Silva et al, out performs their model. Including the extreme values makes the model more accurate. This is owing to the Bayesian approach and the MCMC algorithm. Building a complex model and using an algorithm robust to complexity is better.

Comparing the actual vs predicted PI scores per municipality shows that the MCMC model has a decent fit. Predictions are given as distributions and the plotted values are the mean of each predicted distribution, and hence the model predicts errors in degree of PI.

There are still misfits and errors. As in 2003 in the western states a lesser PI is expected than is seen. In 2010 predicts a high PI in a central state where none exists. Some errors can become 'hits' under the binary prediction approach.
Changing the condition to be a binary yes if the real PI is in the 21% CI of the prediction, answers the question 'is the model right within a given risk tolerance?' as opposed to 'what is the PI?' (at 89% almost all observation are hits, so 21% is chosen to simply demonstrate), the model has better success. This target range can be configured to meet accuracy requirements. Hence the accuracy score at 21% CI is 64%, while at 89% CI it is 100%. The best fitting algorithms for the causal and paper models (MCMC and QA respectively) score 20% hit rate.
Using the model in this way (expanding the hit range) in a sense invalidates accuracy or precision scores, as it enables one to tweak them until they are 'right'. In practice this is a practical usage for decision makers as it allows them to toggle their response based on their aversion to risk. Higher aversion, wider interval.

All these models are asking “what proportion of infected cows relative to all cows tested will be observed conditional on a set of variables.” This is different to asking 'how many infected cows will be observed out of the tested population, conditional on a set of variables'. This latter question is perhaps more useful, and while outside the scope of this paper, it was tested and the MCMC model using the same DAG and La Nina model has a hit rate of 98% (appendix 6). Restating the initial obfuscation inherent in using a proportion (PI) rather than a count.

Conclusion

In the end, using a causal model outperforms a purely predictive model in this situation, even when the latter is modelled on the same Bayesian framework as the causal model. Given a more complex causal model including many outliers caused by La Nina, a more robust algorithm, MCMC, is required to handle model complexity. Indicating that for the disease of bovine fasciolosis, having an understanding of the phenomenon is better than throwing the data at the wall and seeing what sticks.